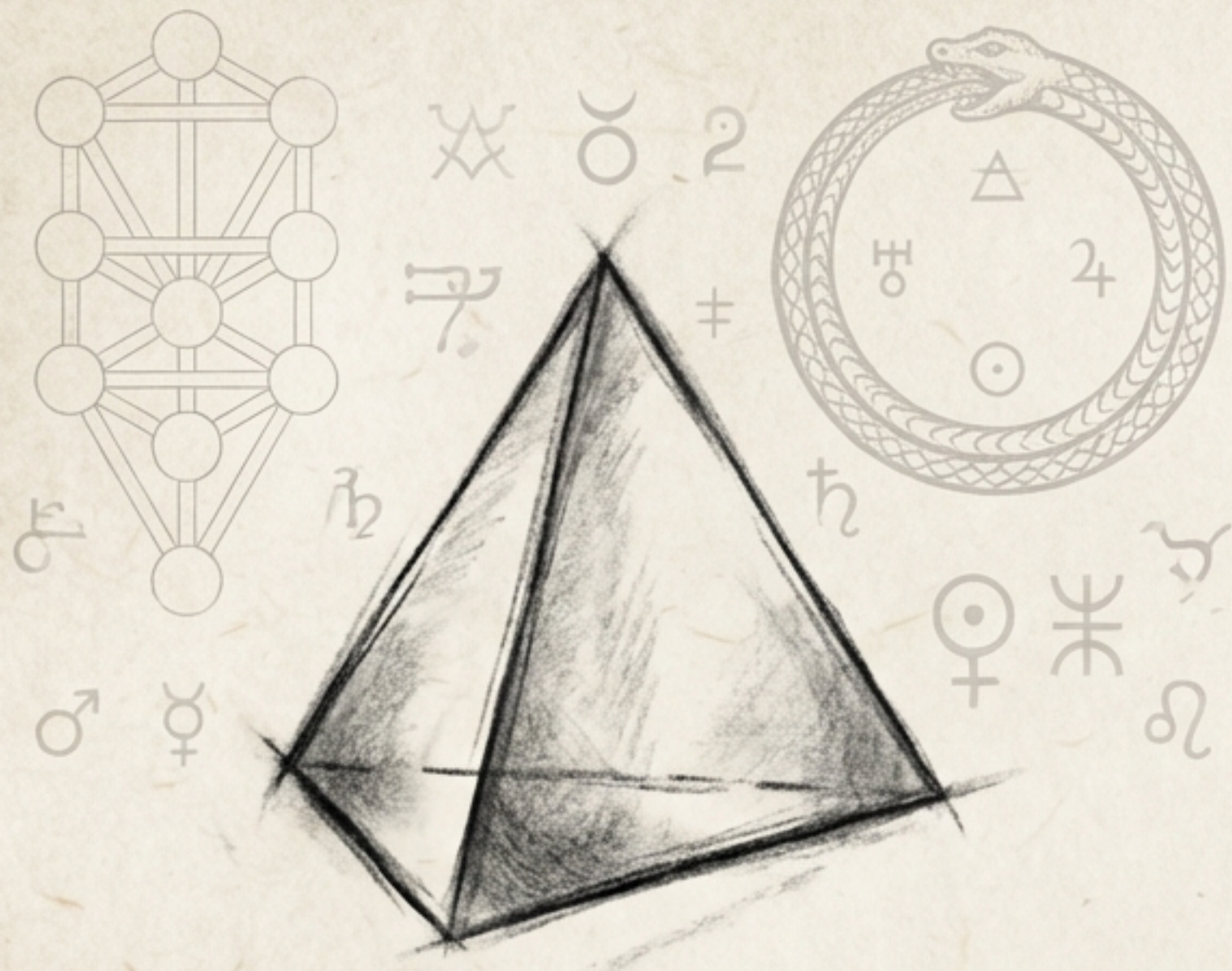


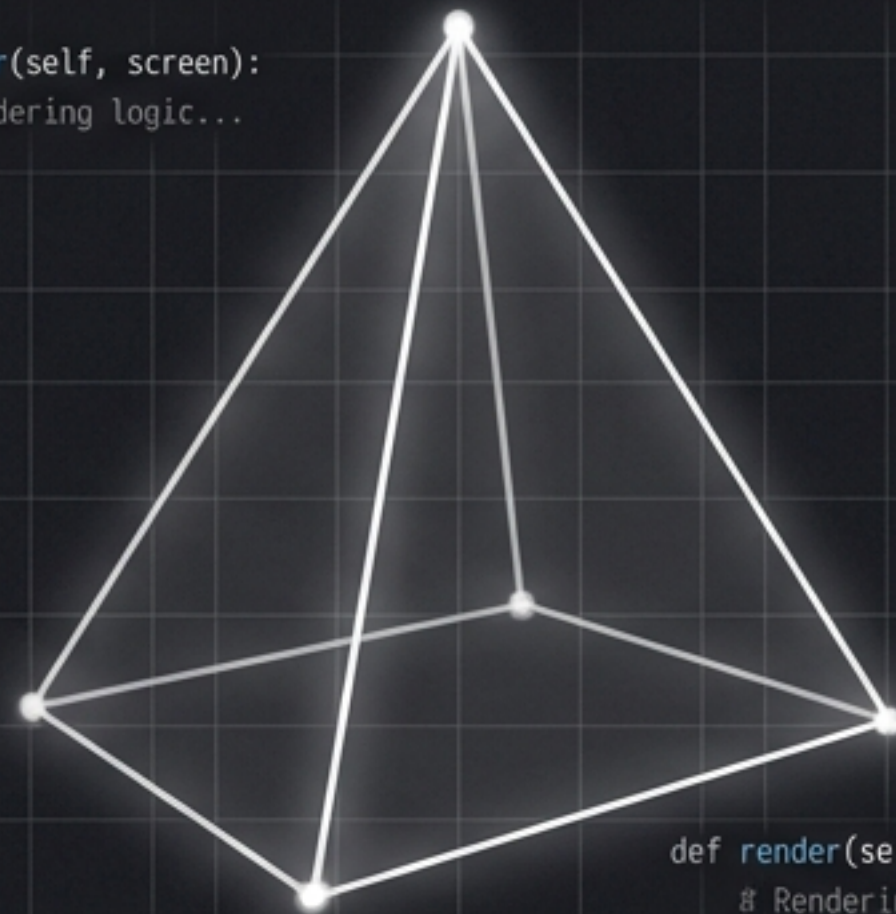


TET~CRAFT: & The Building

An analysis of the dialogue between metaphysical vision and coded reality in a siolated universe.

```
class Tetrahedron:
    def __init__(self):
        self.Edges_NP = np.array([[0, 1], [0, 2], [0, 3], [1, 2], [1, 3], [2, 3]])
        self.Vertices_NP = np.array([[1, 1, 1], [1, -1, -1], [-1, 1, -1], [-1, -1, 1]])
```

```
def render(self, screen):
    # Rendering logic...
```



```
def render(self, screen):
    # Rendering logic...
```

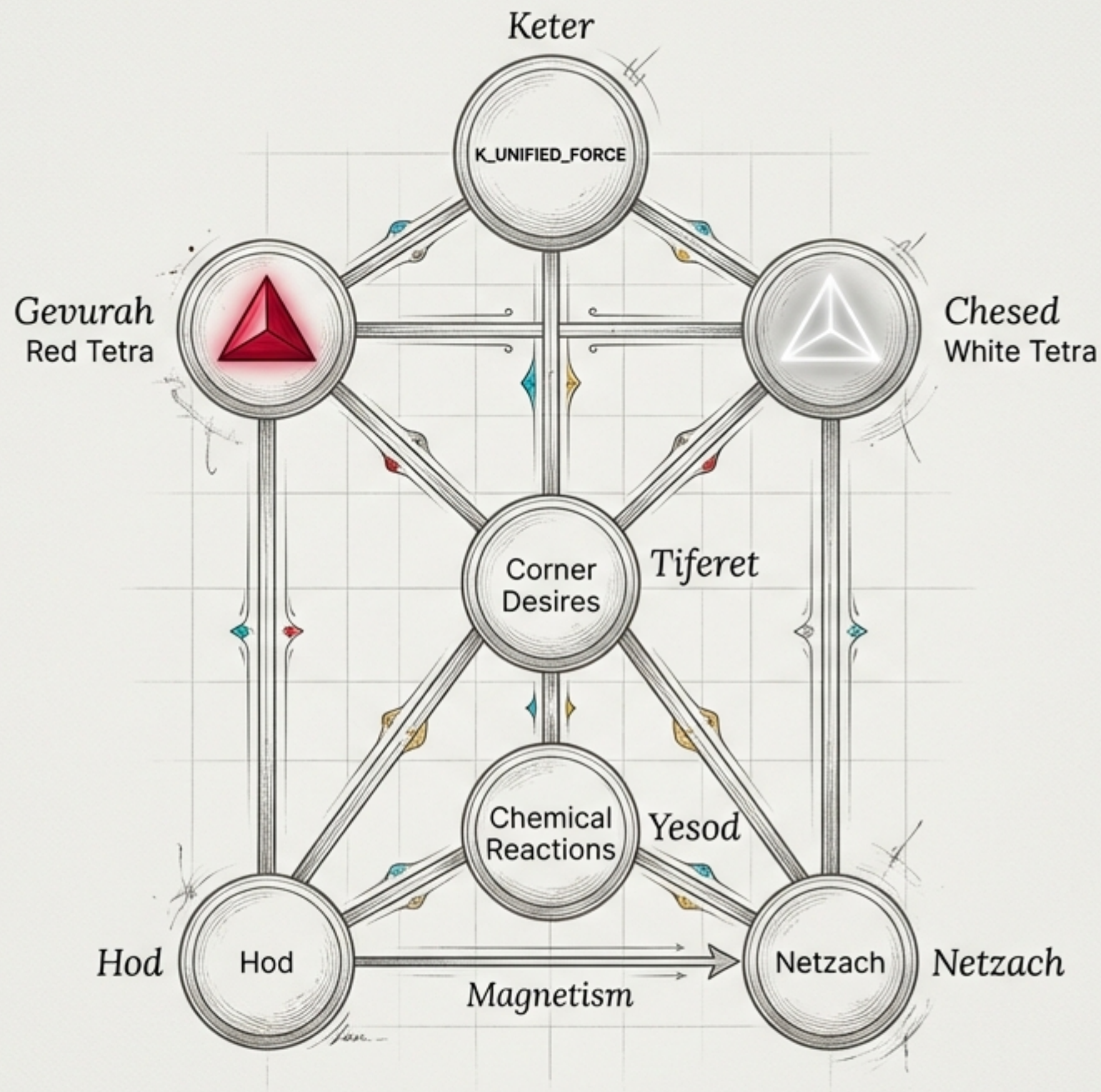

A Universe Born from First Principles



TET~CRAFT is more than a chemistry simulation; it is a "kleiniverse of tetrahedrons that bond into atoms, react, and evolve into complex molecules."

Each tetrahedron, or TET, represents a quantum fact. The simulation's ultimate goal is to model how disconnected facts cohere into understanding.

To simulate understanding formation
from disconnected facts.



The Player as Observer

The player is positioned as “the prime observer missing from quantum theory,” whose act of creation (pressing SPACE) introduces a new “fact (TET) to be misunderstood.”

Kabbalistic Symbolism

The system's forces and colors are mapped to the Sefirot, the divine emanations in Kabbalah.

- White:** Corresponds to *Chesed* (Kindness, Mercy). The color of peace and benevolence.
- Red:** Corresponds to *Gevurah* (Strength, Judgment). The color of intensity, power, and severity.

“The Sefirot are purely spiritual... the colors symbolize the effects and qualities that flow from these higher spiritual roots.”

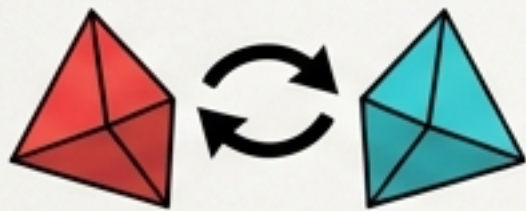
The Building: Geometry is Destiny

PHASE 1: Multi-Face Locking

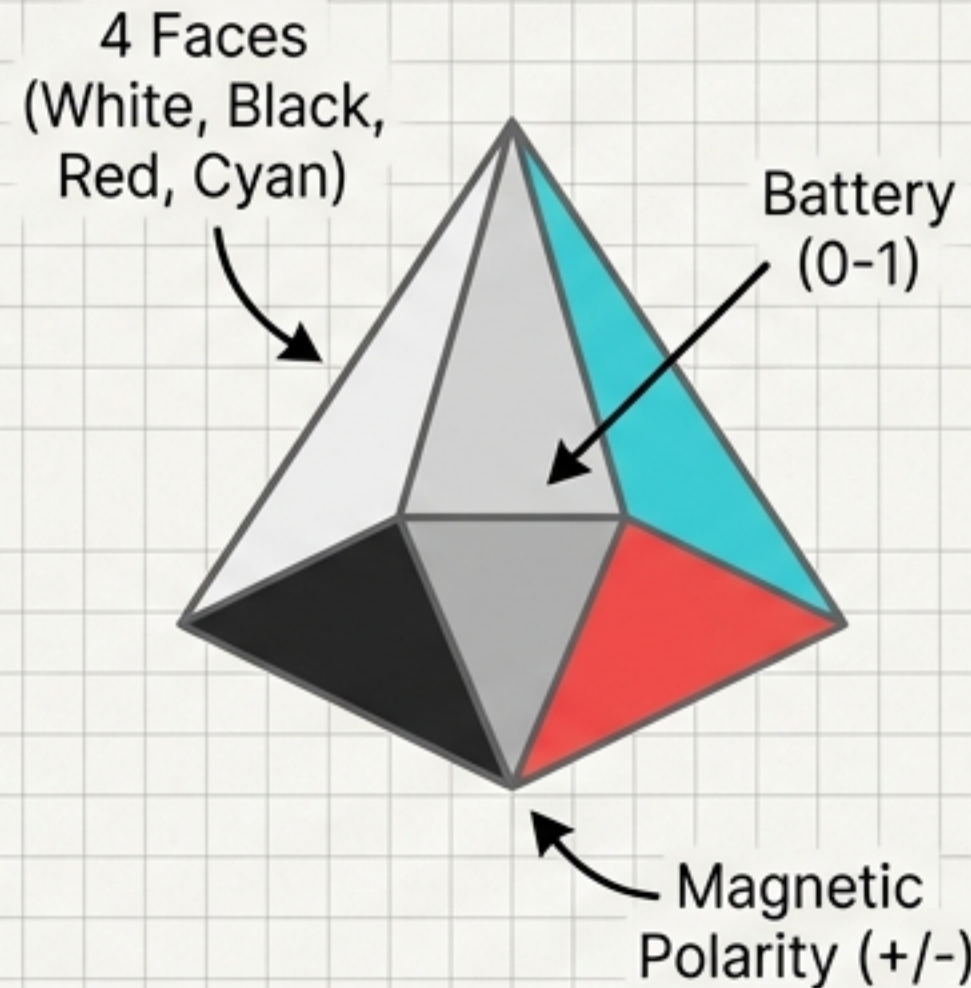


TETs lock faces when vertices connect, defining molecule types.

PHASE 3: Corner Desires



Opposite-colored corners seek each other, driving geometric alignment.

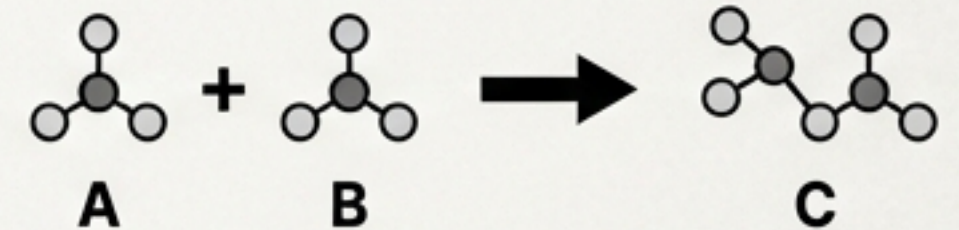


PHASE 2: Battery Oscillation



Bonded pairs exchange energy, creating a constant oscillation.

PHASE 4: Chemical Reactions

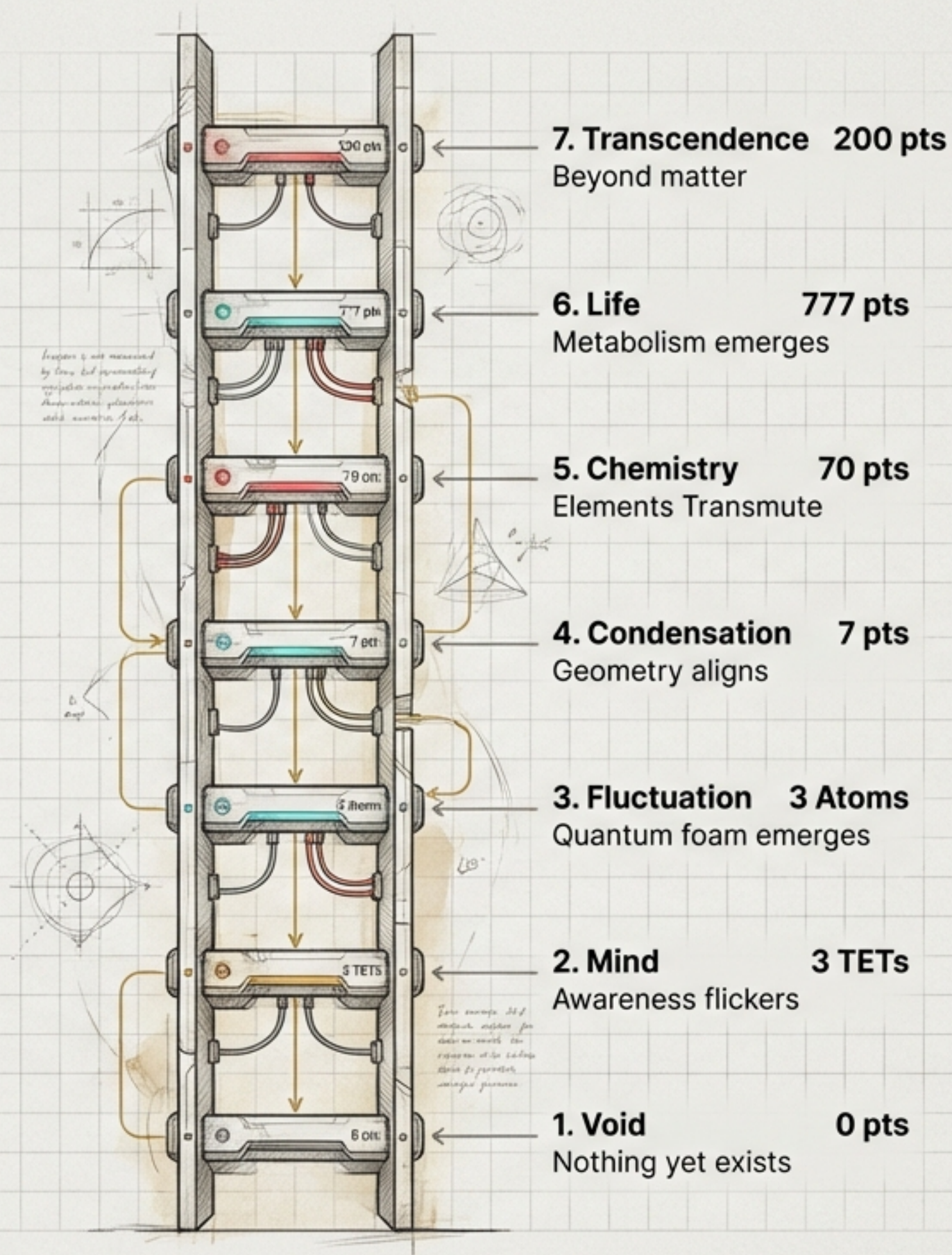


Molecules combine (synthesis) or break down (decomposition).

Core Invariant: Pairing defines elements." A TET's label is meaningless; its physical connections determine its identity.

The Cosmic Ladder: A Measured Ascent

Progress is not measured by time, but by complexity and entropy reduction. The universe gains **Progress Points** through **Synthesis Reactions** and the formation of stable geometric interfaces, ascending through seven distinct Eras.



Systemic Divergence

The architect's inspired sketches

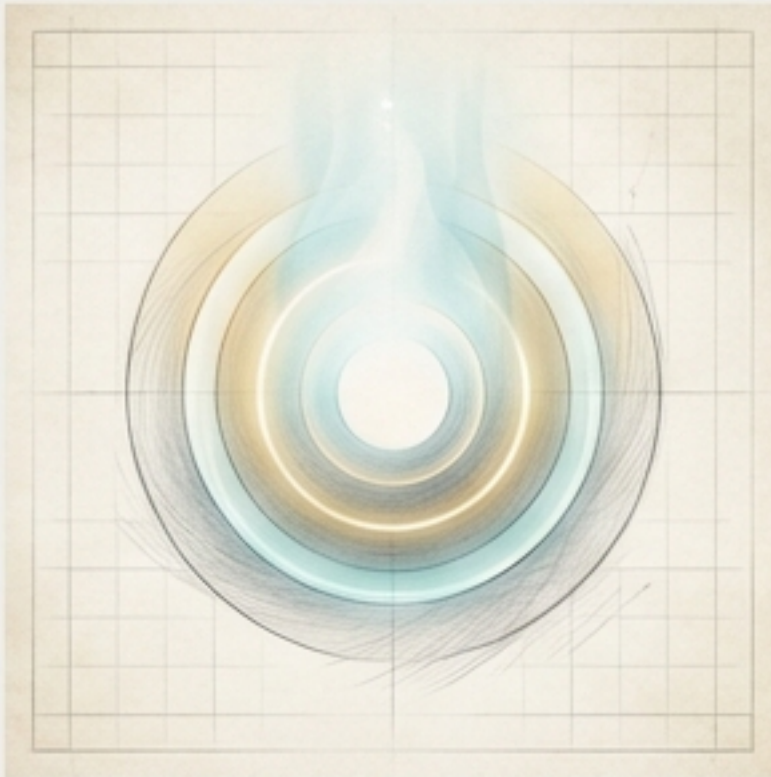
versus

the construction supervisor's reality.

Divergence I: The True Cost of Transcendence

The Blueprint

Source: Cosmic Ladder summary table



Requirement: **200** Progress Points

The Building

Source: Python code (`TechTree` class) & notes

```
class TechTree:  
    STAGES = [  
        ...  
        ('Transcendence', 777, 'Beyond matter'),  
    ]
```

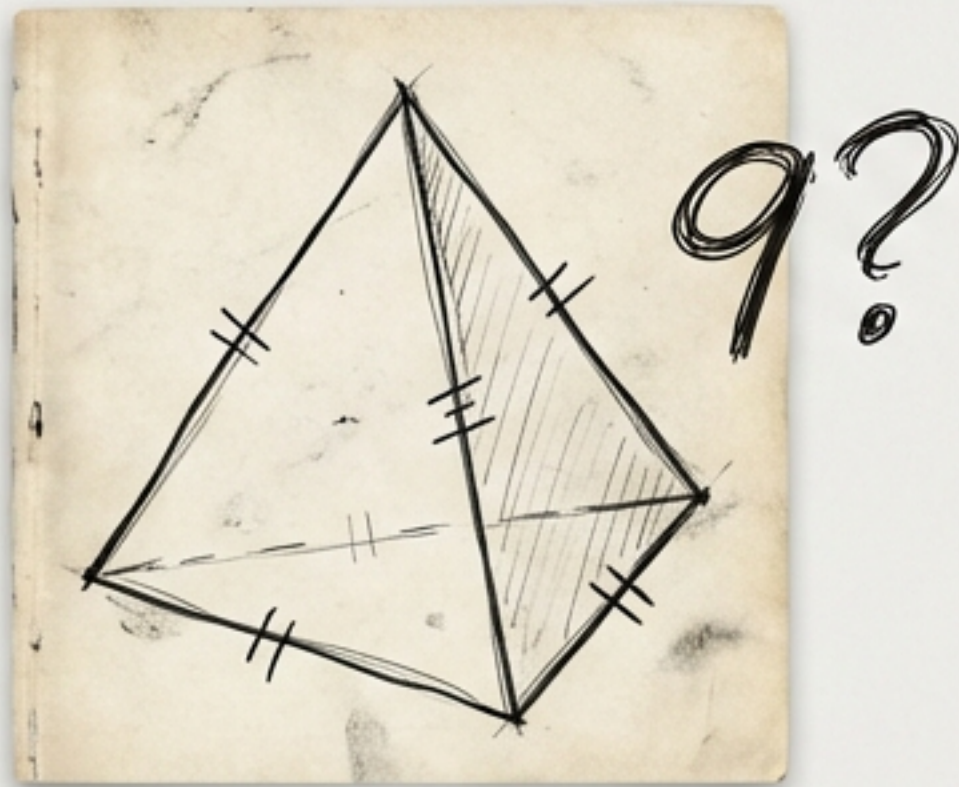
Requirement: **777** Entropy/Progress Points

The implemented reality demands a far greater, perhaps more significant, accumulation of complexity than originally sketched out.

Divergence II: The Shape of Reality

The Blueprint

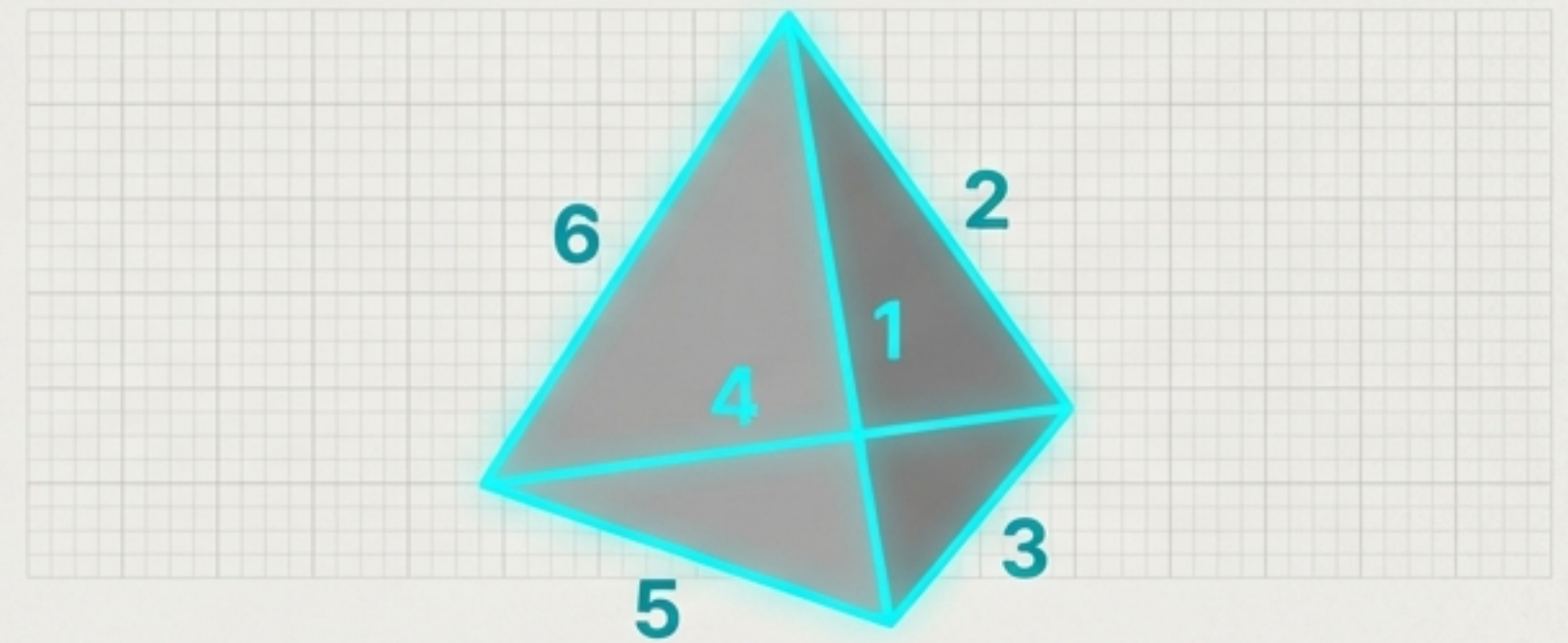
Source: General metaphysical notes



A tetrahedron has '**9? edges**'

The Building

Source: Python physics engine ('EDGES_NP' array)



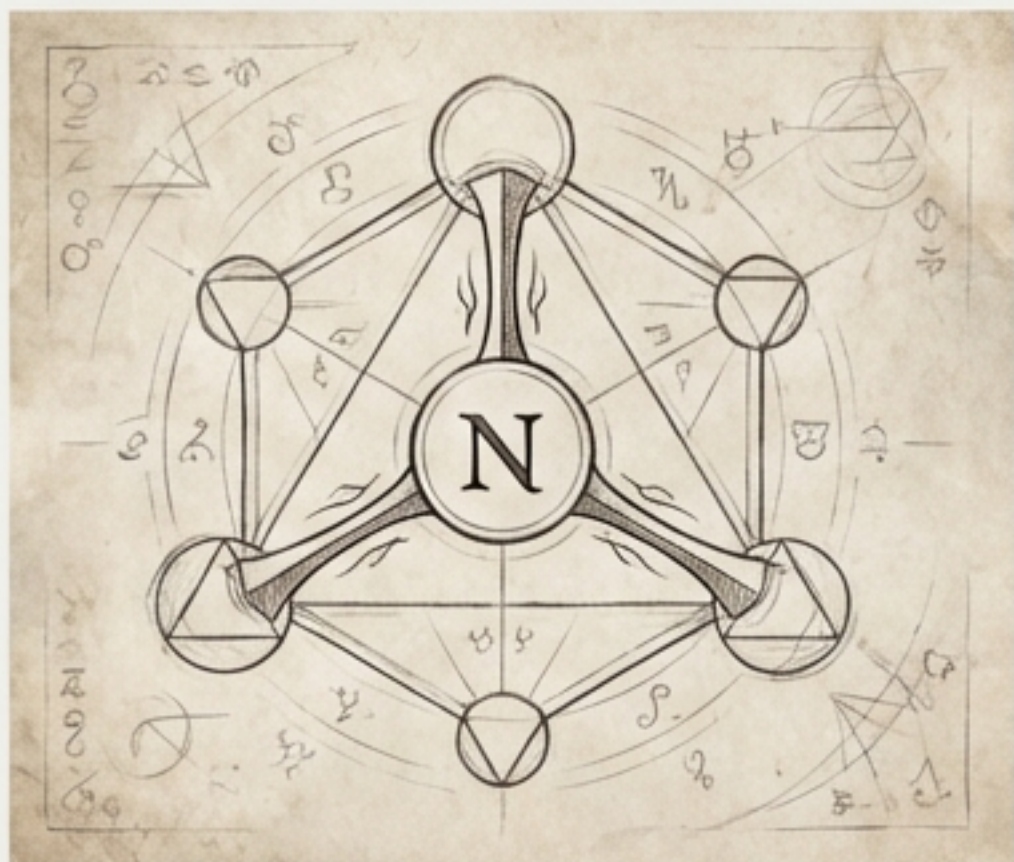
The engine correctly iterates through **6 edges**

The abstract, speculative nature of the vision gives way to the rigid, geometric truths required by the physics engine.

Divergence II: The Constrained Potential of Nitrogen

The Blueprint

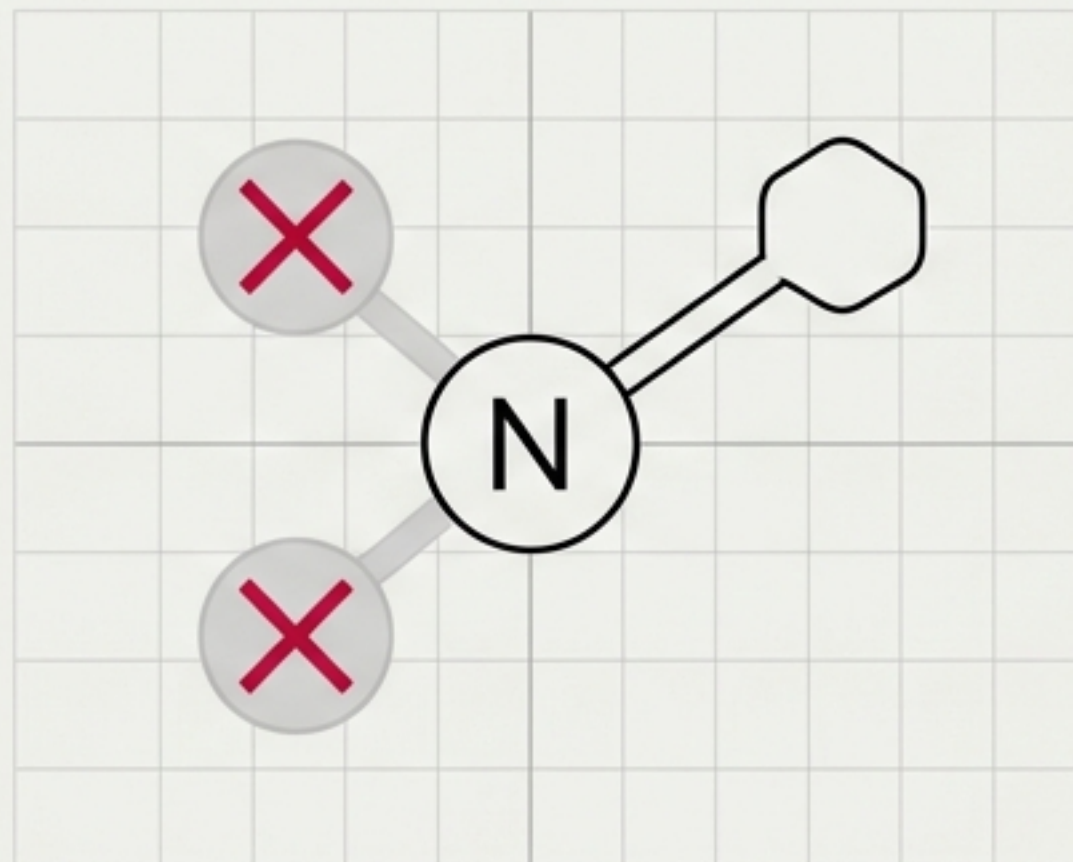
Source: "Chemical Kabbalah" theory



Nitrogen has a **valency of 3**, designed to "support structural complexity."

The Building

Source: Python configuration (``BOND_CAPACITY``)



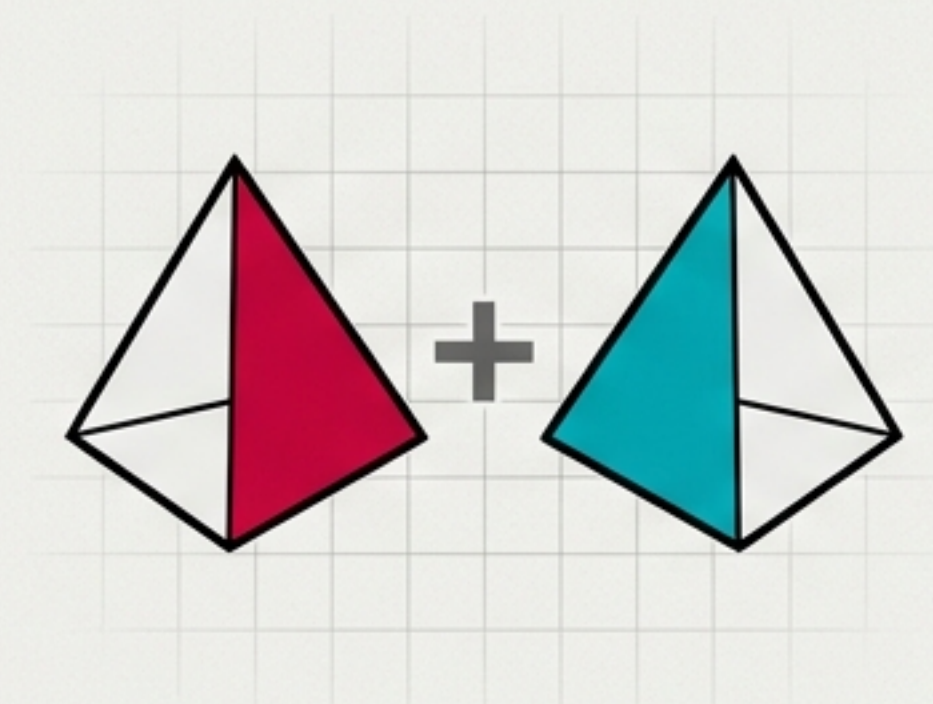
Nitrogen is not listed, falling under the **default bond capacity of 1**.

Ambitious '3-bond' designs are currently held together by a single 'default' bolt.

Synthesis: Where Geometry Forges Matter

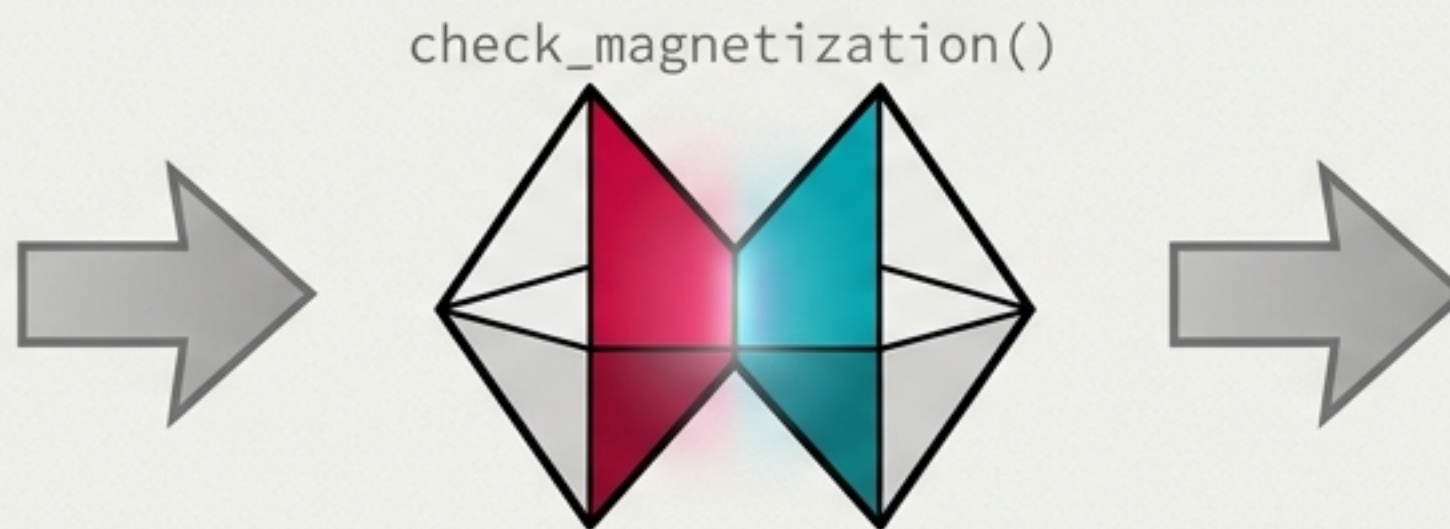
The principle of 'Geometry is Destiny' is most clearly realized in the primordial genesis of Hydrogen.

1. The Rule (Blueprint)



The Interface Chemistry Table dictates that a physical connection of **Cyan** ↔ **Red** faces transmutes matter into Hydrogen (H).

2. The Action (Building)



The Python `'check_magnetization'` function enforces this rule.

3. The Reward



The formation provides a precise energy refill of **0.001** to the participating TETs, a tangible reward for following the geometric law.

From Random Collisions to a Story of Chemistry

Analogy

Bot Intelligence is explained through the metaphor of **"a room full of heat-seeking drones in total darkness."**

- * Advanced Mechanics: Are the 'infrared sensors' that allow the drones to find the 'warmth' of energy-rich molecules.
- * The BotMind: Is the 'internal logbook' where the drone records what it finds.

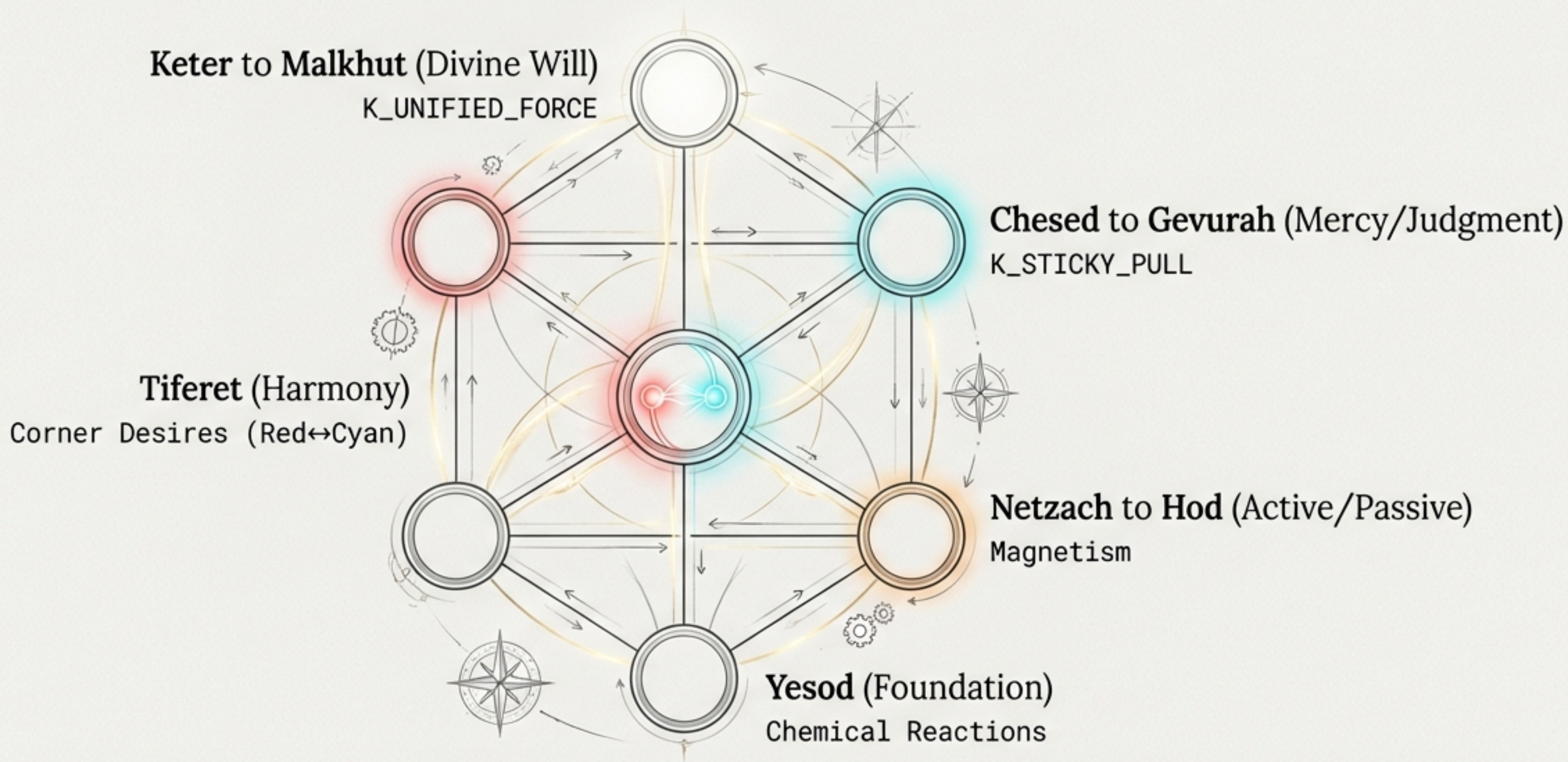


Key Insight:

This process transforms **"a room of random collisions into a documented story of chemistry,"** demonstrating how the rigid **'Building'** successfully generates the emergent narrative envisioned in the **'Blueprint.'**

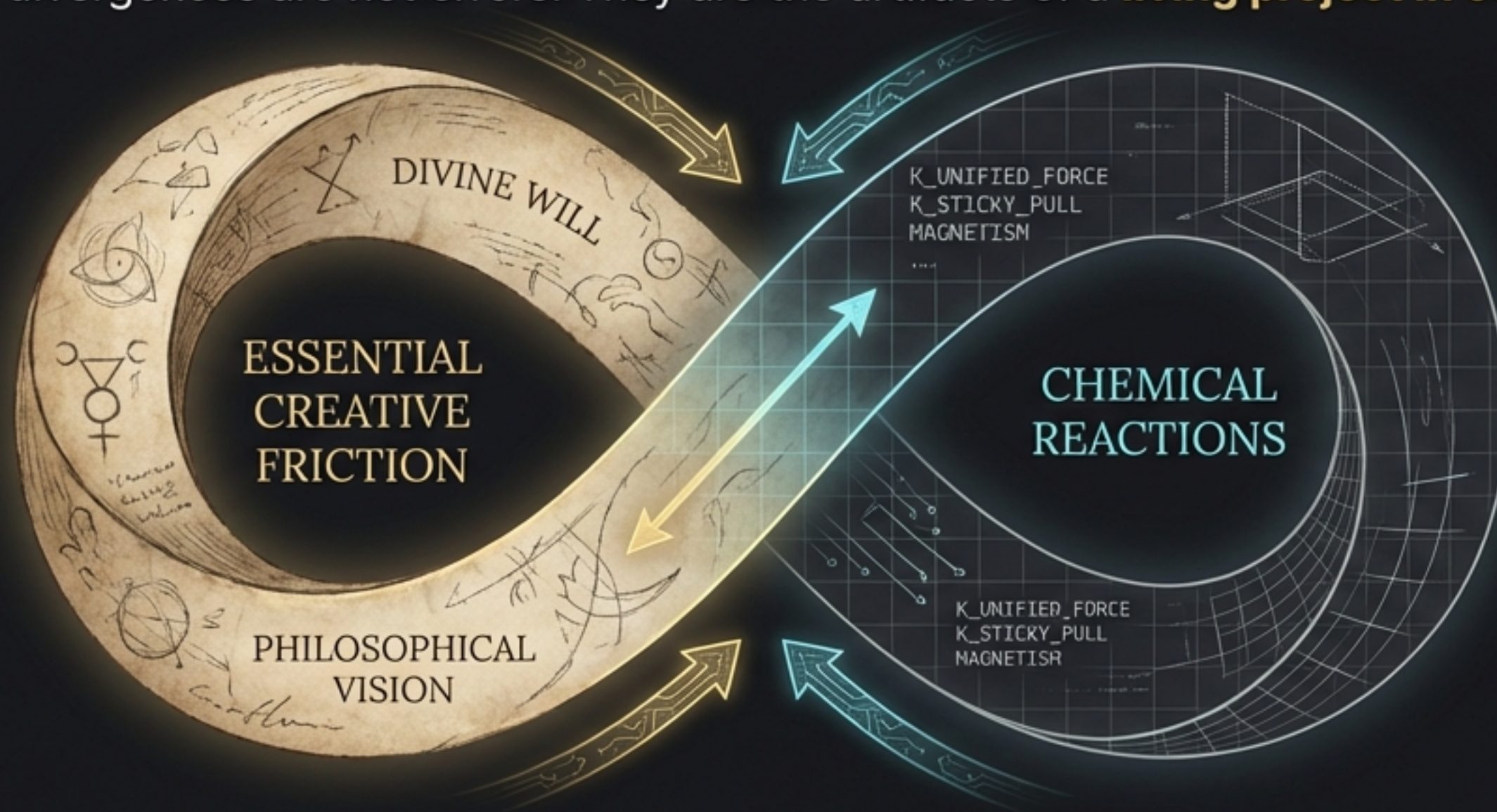
A Dynamic Tree of Life, Written in Code

The Sefirotic correspondences are not merely flavor text; they are the soul of the physics engine.
Each major force constant in the code is a direct implementation of a **Kabbalistic principle**.



Synthesis as a Feature, Not a Bug

The documented divergences are not errors. They are the artifacts of a **living project in constant evolution**.



They reveal the essential, **creative friction** between a boundless philosophical **vision**-endeavour and the pragmatic realities execution.

This dialogue is what drives **TET~CRAFT** forward.

TET~CRAFT v5.3D: System Specifications

Overview

Program

TET~CRAFT: The Fourth Temple

Version

v5.3D

Concept

Emergent Quantum Chemistry & Meta Physics Simulation

Core Invariants

1. Pairing defines elements.

2. Physics $\neq$ Chemistry $\neq$ Gameplay ... yet

3. Catalysts cannot transmute without mediators.

4. Low energy causes positional drift...

Key Systems

TET Properties

4 faces, Battery, Polarity

Chemistry

4 Phases, >100 molecule types

Physics

Magnetic bonding, Singularity pull

Bot Intelligence

Automates exploration, generates thoughts



“God is the fractal to which we are each a frame...
The player [is] the prime observer missing from quantum theory.
Pressing space creates a new fact(TET) to be misunderstood.”